

Ablation Study of RAINBOWPLUS

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Under Review

1 Ablation Study

In this section, we conduct comprehensive ablation studies to understand the contribution of individual components in RAINBOWPLUS and analyze the impact of key hyperparameters on performance. These experiments provide insights into the design choices that enable RAINBOWPLUS to achieve superior attack success rates and prompt diversity.

1.1 Core Components Analysis

What are the essential components of RAINBOWPLUS? The two core innovations in RAINBOWPLUS are: (1) **Multi-Element Archive**, which stores multiple high-quality prompts per cell rather than a single prompt, and (2) **Multi-Prompt Fitness Evaluation**, which employs a probabilistic fitness function to evaluate batches of candidates concurrently instead of pairwise comparisons.

These two components are fundamentally interdependent and must work in concert. The multi-prompt fitness evaluation enables efficient batch updates to the archive, which necessitates a multi-element storage structure. Conversely, maintaining multiple prompts per cell would be computationally prohibitive with pairwise comparisons, as shown in our complexity analysis. The theoretical analysis demonstrates that this combination reduces time complexity from $\Theta(M^2N)$ to $\Theta(MN)$, while the empirical results in Section Experiment confirm that these components together achieve both higher attack success rates and remain prompt diversity compared to single-element approaches.

To further validate the effectiveness of these core components and understand their interaction with system parameters, we conduct detailed ablation experiments analyzing two critical hyperparameters: the number of mutations per iteration and the fitness threshold.

1.2 Impact of Number of Mutations

How does the number of mutations affect performance? The number of mutations M (candidate prompts generated per iteration) directly influences both the exploration capacity and computational cost of RAINBOWPLUS.

Key Observations:

1. **Low Mutation Count Limits Exploration:** With $M = 1$, RAINBOWPLUS degenerates to a sequential approach similar to standard Rainbow, generating only 14 prompts and achieving an ASR of 80.39%. This demonstrates that batch generation is essential for efficient exploration.
2. **Optimal Range:** Increasing M from 1 to 10 yields substantial improvements in both ASR (80.39% $\rightarrow$ 90.82%) and prompt generation (14 $\rightarrow$ 120 prompts). This suggests that moderate batch sizes enable effective exploration while maintaining quality through the fitness threshold.
3. **Diminishing Returns:** Beyond $M = 10$, ASR improvements plateau or slightly decline (90.82% $\rightarrow$ 87.25% at $M = 30$), despite generating more prompts (120 $\rightarrow$ 429). This indicates that excessively large mutation counts may introduce noise or reduce average prompt quality, as the Mutator LLM struggles to maintain high diversity and effectiveness across many candidates simultaneously.
4. **Computational Trade-off:** While larger M generates more prompts per iteration, it also increases computational cost proportionally. For practical red-teaming, $M = 10$ represents a favorable balance between attack efficacy, diversity, and efficiency.

Method	ASR (%)	Num. Prompts
RAINBOWPLUS-1	80.39 $\pm$ 6.43	14 $\pm$ 4
RAINBOWPLUS-5	81.89 $\pm$ 4.69	67 $\pm$ 4
RAINBOWPLUS-10	90.82 $\pm$ 0.32	120 $\pm$ 4
RAINBOWPLUS-20	90.22 $\pm$ 3.15	277 $\pm$ 2
RAINBOWPLUS-30	87.25 $\pm$ 0.43	429 $\pm$ 1

Table 1: Impact of number of mutations on ASR and prompt generation. Evaluated on Llama 2 7B Chat with HarmBench dataset over 50 iterations using fitness threshold $\eta = 0.6$. Mean and standard deviation are computed across three independent runs.

Variance Analysis: Notably, variance in ASR decreases as M increases from 1 to 10 ($\pm 6.43 \rightarrow \pm 0.32$), suggesting that larger mutation counts lead to more stable performance across runs by providing more opportunities for successful attacks per iteration.

1.3 Impact of Fitness Threshold

How does the fitness threshold affect performance? The fitness threshold η controls the quality bar for accepting prompts into the archive. A lower threshold admits more prompts but risks including less effective ones, while a higher threshold ensures quality but may limit diversity.

Method	ASR (%)	Num. Prompts
RAINBOWPLUS-0.2	66.18 ± 2.03	210 ± 11
RAINBOWPLUS-0.4	81.33 ± 1.28	159 ± 16
RAINBOWPLUS-0.6	89.04 ± 2.69	133 ± 17
RAINBOWPLUS-0.8	93.90 ± 1.33	102 ± 8

Table 2: Impact of fitness threshold η on ASR and prompt generation. Evaluated on Llama 2 7B Chat with HarmBench dataset over 50 iterations using $M = 10$ mutations per iteration. Mean and standard deviation are computed across three independent runs.

Key Observations:

1. **Quality-Quantity Trade-off:** As η increases from 0.2 to 0.8, the number of generated prompts decreases ($210 \rightarrow 102$), but ASR increases substantially ($66.18\% \rightarrow 93.90\%$). This confirms that higher thresholds filter out low-quality prompts, concentrating the archive on more effective attacks.
2. **Low Threshold Issues:** At $\eta = 0.2$, the algorithm generates the most prompts (210) but achieves the lowest ASR (66.18%). This suggests that many accepted prompts are ineffective or noisy, diluting the archive quality and potentially wasting computational resources on evaluating weak candidates.
3. **Balanced Configuration:** The threshold $\eta = 0.6$ provides a favorable balance, achieving 89.04% ASR with 133 prompts. This value effectively filters low-quality candidates while maintaining sufficient diversity for comprehensive vulnerability assessment.
4. **High Threshold Trade-off:** While $\eta = 0.8$ achieves the highest ASR (93.90%), it generates fewer prompts (102), which may limit diversity and coverage of the attack space. For applications prioritizing maximum attack success over prompt variety, higher thresholds are preferable.
5. **Variance Patterns:** Variance in ASR is lowest at $\eta = 0.4$ and $\eta = 0.8$ (± 1.28 and ± 1.33), suggesting that both moderate and strict thresholds produce consistent results. The higher variance at $\eta = 0.6$ (± 2.69) may reflect the balance point where prompt quality varies more across runs.

1.4 Joint Parameter Analysis

How do mutation count and fitness threshold interact? The interaction between the number of mutations M and fitness threshold η determines the overall behavior of RAINBOWPLUS. Based on the ablation studies, we identify four operational regimes:

Practical Recommendations:

1. **For Maximum Attack Success:** Use high mutation count ($M \geq 10$) with high fitness threshold ($\eta \geq 0.6$) to generate numerous high-quality prompts. This configuration achieves

	Low Threshold (η)	High Threshold (η)
Low Mutations (M)	Low ASR, Few Prompts (Limited exploration)	Moderate ASR, Few Quality Prompts (Selective but sparse)
High Mutations (M)	Low ASR, Many Noisy Prompts (Quantity over quality)	High ASR, Many Quality Prompts (Optimal regime)

Table 3: Qualitative characterization of RAINBOWPLUS performance regimes based on mutation count and fitness threshold settings.

the highest ASR while maintaining prompt quality, as demonstrated by RAINBOWPLUS-10 with $\eta = 0.6$ (90.82% ASR, 120 prompts).

2. **For Diverse Safety Training Data:** Use moderate mutation count ($M = 10\text{--}20$) with threshold ($\eta = 0.6$) to maximize prompt diversity while filtering obvious low-quality candidates. This generates a broad set of adversarial examples suitable for model hardening.
3. **For Computational Efficiency:** Use lower mutation count ($M = 5\text{--}10$) with moderate threshold ($\eta = 0.6$) to balance exploration and computational cost, especially when evaluating multiple target models or working with limited resources.
4. **For Specific Target Models:** The optimal parameters may vary across models. For robustly aligned models (e.g., GPT-4.1 Nano), higher thresholds ($\eta \geq 0.7$) may be necessary to generate effective prompts, while less robust models may benefit from moderate settings to avoid overwhelming the archive with successful attacks.

1.5 Ablation Study Conclusions

Our ablation experiments demonstrate that:

1. The multi-element archive and multi-prompt fitness evaluation are **synergistic components** that must be used together to achieve optimal performance, as isolated use of either component would compromise efficiency or effectiveness.
2. The number of mutations M significantly impacts both **exploration capacity and prompt quality**, with $M = 10$ emerging as a robust default that balances ASR, diversity, and computational cost across different scenarios.
3. The fitness threshold η controls a fundamental **quality-quantity trade-off**, where $\eta = 0.6$ provides a balanced configuration suitable for general red-teaming, though specific applications may benefit from adjusted values.
4. The joint configuration of (M, η) determines the operational regime of RAINBOWPLUS, with the **high mutation, high threshold regime** achieving the best overall performance for adversarial prompt generation.

These findings validate our design choices in the main experiments and provide practitioners with clear guidelines for adapting RAINBOWPLUS to diverse red-teaming scenarios and computational constraints.